Improving Genome Recovery Through Similarity-Guided Co-assembly

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1. HYPOTHESIS

Co-assembly of multiple metagenomes improves genome recovery, especially for rare microbes. Because co-assembly is computationally demanding, samples must be grouped strategically.

Here, we test different **k-mer-based clustering** strategies to identify suitable metagenome groups for efficient co-assembly.

2. DATASETS

Mixed Set

25 metagenomes of 5 different types

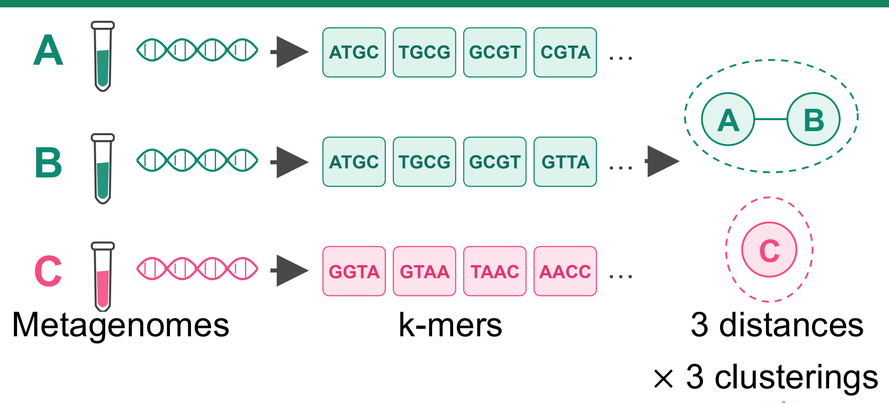
Bucket sewage | Time-series sewage | Human feces | Cat feces | Soil

European sewage time series

278 metagenomes [1]

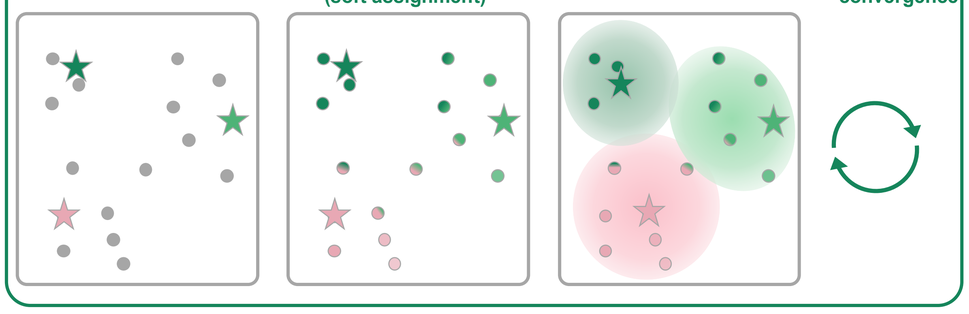
Copenhagen x3 WWTPs | Rotterdam | Bologna | Budapest | Rome

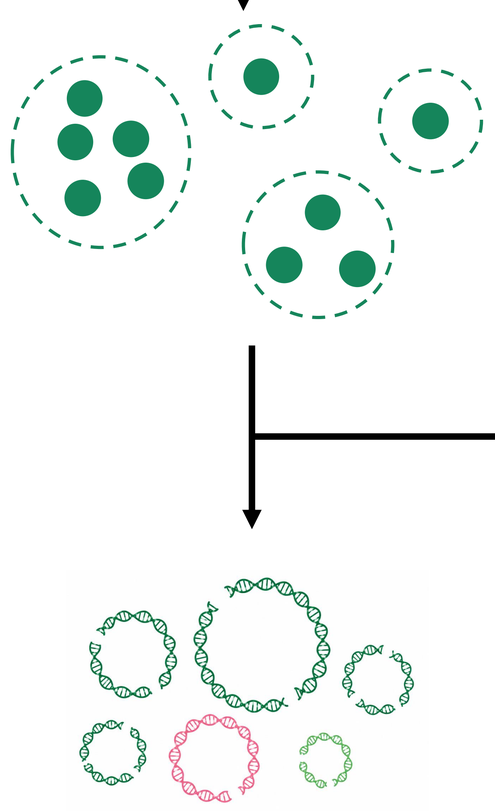
4. K-MER-BASED CLUSTERING



Distances: Mash, Sourmash, Sourmash abundance

Clusterings: DBSCAN, k-medoids, **fuzzy k-medoids**





5. PROCESSING OF CLUSTERS

- Co-assembly *(MetaSPAdes & MEGAHIT)*
- Binning *(MetaBAT2)*
- Quality assessment *(CheckM2)*
- Dereplication *(dRep to 95% ANI and at least medium quality)*
- Taxonomic annotation *(GTDB-Tk)*

→ Metagenome-Assembled Genomes (**MAGs**)

6. RESULTS

Table1: MAGs recovered by co-assembly strategies and individual assembly. Mixed total: n = 202. Sewage total: n=2,072. Sewage individual = single-sample assembly [1]. DBSCAN for sewage resulted in one cluster, therefore not available.

		Mixed set (25)			Sewage set (278)		
		Species	% of all species	Mean N50 (kb)	Species	% of all species	Mean N50 (kb)
Mash	KM	169	84%	15.3	1423	69%	19.6
	FKM	169	84%	15.3	1591	77%	19.9
	DBSCAN	172	85%	13.2	–	–	–
Sourmash Jaccard	KM	161	80%	17.6	1429	69%	20.4
	FKM	172	85%	15.2	1644	79%	21.1
	DBSCAN	169	84%	15.3	–	–	–
Sourmash abundance	KM	155	77%	17.5	1252	60%	20.6
	FKM	161	80%	14.6	1451	70%	19.7
	DBSCAN	159	79%	15.0	–	–	–
Individual		113	56%	30.1	925	–	–

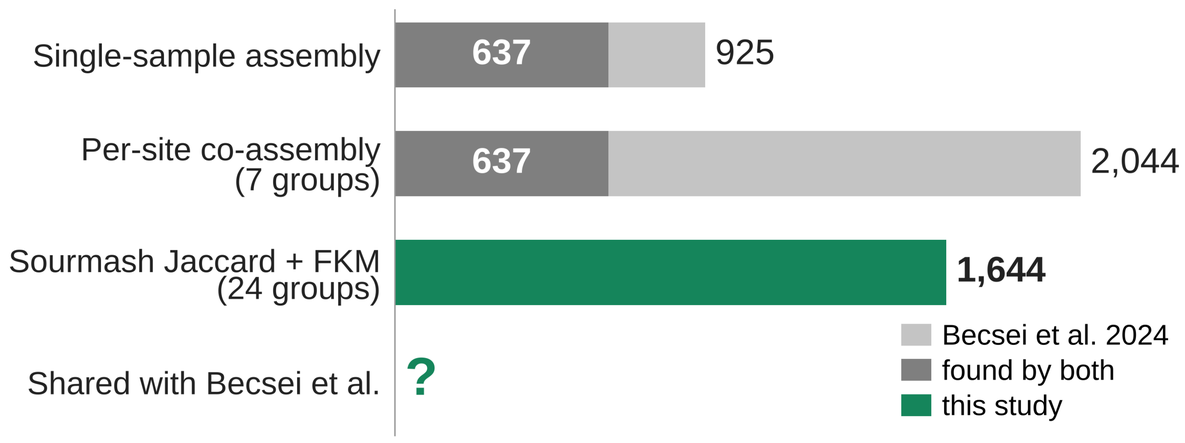


Figure1: Species recovered from the sewage set (278 samples). Species-level medium-quality MAGs from single-sample and per-site co-assembly in [1], compared with Sourmash Jaccard + FKM (this study). The overlap between this study and [1] has not been determined yet.

- Co-assembly increased recovery of MAGs **up to 52% and 78%** compared to individual assembly for the 2 datasets, respectively.
- Sourmash Jaccard + FKM results in highest number of recovered MAGs in both datasets (and highest N50 for sewage).
- Whether this is driven by cluster size or by sample similarity remains to be investigated.

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References:

[1] Becsei, A. *et al*. Time-series sewage metagenomics distinguishes seasonal, human-derived and environmental microbial communities potentially allowing source-attributed surveillance. *Nat. Commun.* **15**, 7551 (2024).



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